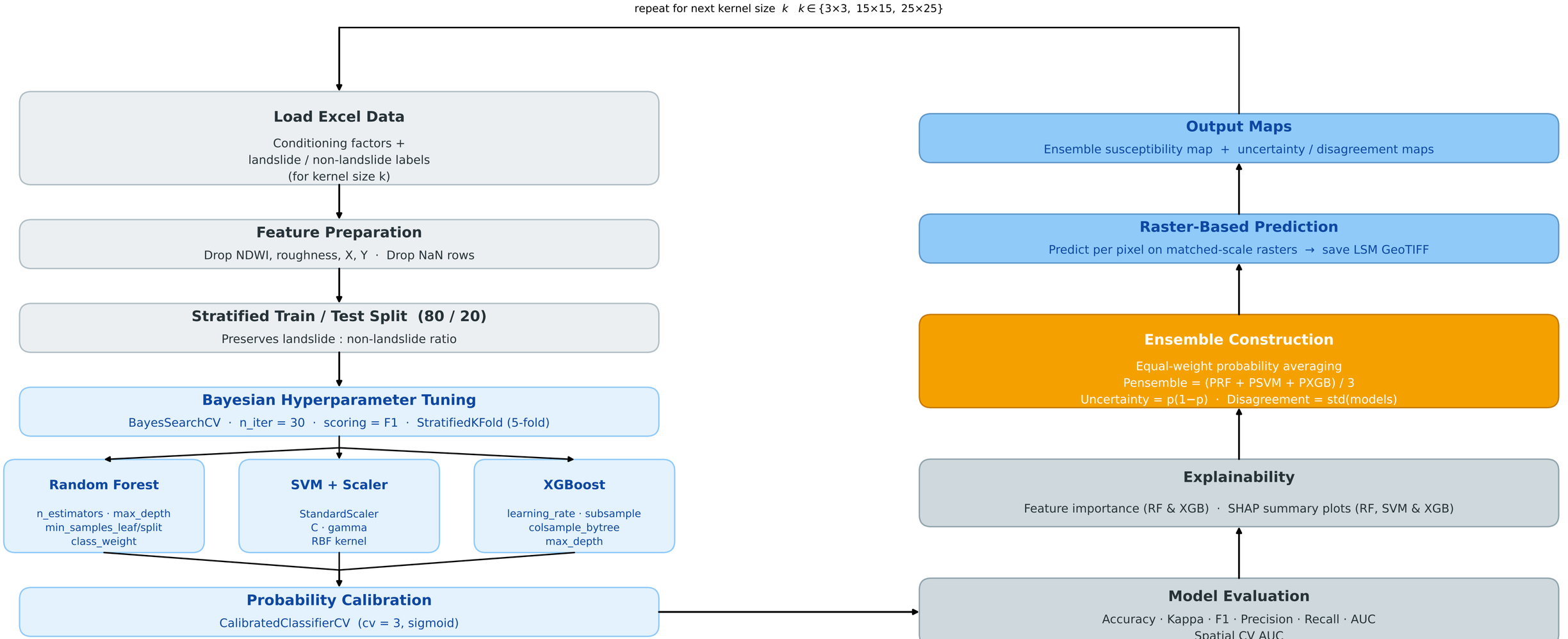
repeat for next kernel size k $k \in \{3\times3,\ 15\times15,\ 25\times25\}$
Load Excel Data
Conditioning factors +
landslide / non-landslide labels
(for kernel size k)
Feature Preparation
Drop NDWI, roughness, X, Y · Drop NaN rows
Stratified Train / Test Split (80 / 20)
Preserves landslide : non-landslide ratio
Bayesian Hyperparameter Tuning
BayesSearchCV · n_iter = 30 · scoring = F1 · StratifiedKFold (5-fold)
Random Forest
n_estimators · max_depth
min_samples_leaf/split
class_weight
SVM + Scaler
StandardScaler
C · gamma
RBF kernel
XGBoost
learning_rate · subsample
colsample_bytree
max_depth
Probability Calibration
CalibratedClassifierCV (cv = 3, sigmoid)
Model Evaluation
Accuracy · Kappa · F1 · Precision · Recall · AUC
Spatial CV AUC
Explainability
Feature importance (RF & XGB) · SHAP summary plots (RF, SVM & XGB)
Ensemble Construction
Equal-weight probability averaging
Pensemble = (PRF + PSVM + PXGB) / 3
Uncertainty = p(1−p) · Disagreement = std(models)
Raster-Based Prediction
Predict per pixel on matched-scale rasters → save LSM GeoTIFF
Output Maps
Ensemble susceptibility map + uncertainty / disagreement maps